

SRN Solar PV – Khaldiya District

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ENG208 : Introduction to Sustainable Energy

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LOGO



**SRN SOLAR
PV
SYSTEM
KHALDIYA
DISTRICT**

INTRODUCTION

- Kuwait relies heavily on fossil fuels for electricity generation.
- High temperatures increase electricity demand, especially for air conditioning.
- This leads to high energy consumption and environmental impact.
- Renewable energy, particularly solar energy, is a sustainable solution.
- This project focuses on applying solar PV systems in Al-Khalidiya district.



DISTRICT DESCRIPTION

- **Al-Khalidiya is a developed residential area in Kuwait City.**
- **Population: approximately 19,000–20,000 residents.**
- **Includes residential buildings, services, and educational institutions.**
- **High electricity consumption due to:**
 - **Households**
 - **Commercial activities**
 - **Air conditioning systems**
- **Suitable for solar energy implementation**



ENERGY DEMAND ESTIMATION

- Population $\approx$ 19,300 residents
- Average consumption in Kuwait = 15,000 kWh/person/year
- Total demand calculated using:
Demand = Population $\times$ Consumption
- Total annual demand:
 - 289,500,000 kWh/year
 - $\approx$ 289.5 GWh/year
 -
- This value is used to design the solar PV system



RENEWABLE ENERGY DESIGN

High Solar Potential:

- Strong sunlight year-round; solar irradiance: 5.5 kWh/m²/day (Al-Shammari & Al-Mahdi, 2022)
- According to NREL (2023), a 128 MW PV system generates 218 GWh annually, meeting 50% of district demand.

Efficiency Maximization:

- Use monocrystalline panels (higher absorption than polycrystalline)
- Apply anti-reflection coating and bifacial panels (reduces 30% reflection loss)
- Integrate ground-mounted trackers (panels follow sunlight)

Available Infrastructure:

- Rooftops: 12 schools, 15 mosques, co-ops, government offices, residences, which equals 88 MW capacity
- Solar carports: Estimated 1.24 km² > Required 1.15 km² which is sufficient

Installed Capacity Required:

- Annual renewable energy need: 144,750,000 kWh (50% of total consumption)
- Required installed capacity: 85 MW
- Land area needed: 10,000 m² per MW which mean 850,000 m² (distributed across rooftops & parking shade structures)

ENVIRONMENTAL AND PUBLIC HEALTH BENEFITS

Carbon Reduction:

- 85 MW system generates 144,750,000 kWh/year of clean electricity
- Avoids 72,375 metric tons CO₂/year (U.S. EPA, 2023)
- Equivalent to planting 1.2 million trees and allowing them to grow for a decade

Co-Pollutant Reduction (National Research Council, 2010):

- Sulfur dioxides (SO_x)
- Nitrogen oxides (NO_x)
- Fine particulate matter (PM_{2.5})

Public Health Benefits (Munoz et al., 2021):

- Fewer hospital admissions for respiratory diseases
- Lower rates of asthma and other respiratory illnesses
- Improved local air quality
- Greater energy independence for Khaldiya district

SOCIAL IMPACT

- Improves community well-being by providing a cleaner and healthier environment.
- Improves public health by reducing air pollution and harmful emissions.
- Increases awareness of renewable energy and sustainable practices.



ECONOMIC IMPACT

- Reduces long-term electricity generation and operating costs.
- Creates employment opportunities, especially for AASU majors such as Energy Systems Engineering and Environmental Engineering.
- Supports Kuwait's economic diversification and reduces dependence on fossil fuels.

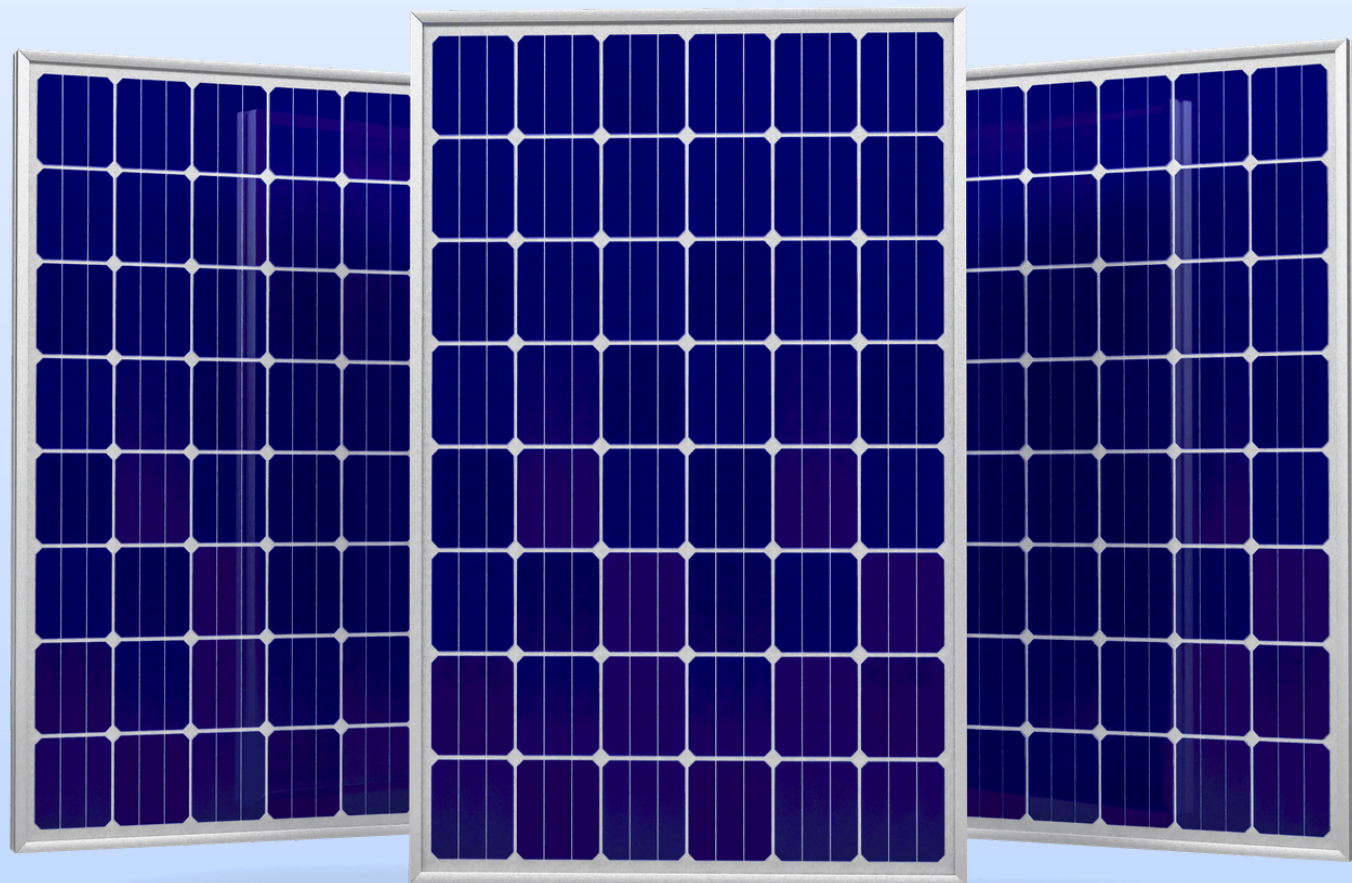


HDI IMPACT

- Improves the health index through cleaner air and better living conditions, contributing to an estimated Life Expectancy Index of 0.892.
- Supports the education index through reliable electricity for schools and digital learning systems, with an estimated Education Index of 0.856.
- Enhances the income index by creating jobs and improving economic opportunities, with an estimated Income Index of 0.945, resulting in an overall HDI of 0.897 for Khaldiya.



KEY POINTS FROM DR.MORSI AHMED INTERVIEW



- Renewable energy transition in the Gulf region requires major infrastructure changes because current systems mainly depend on oil and gas.
- Dust accumulation and high temperatures in Kuwait can reduce the efficiency of solar panels and require regular maintenance.
- Although renewable energy projects need high initial investment, they provide long-term environmental and economic benefits.
- Advanced materials such as Perovskites can improve solar panel efficiency and support future clean energy projects.

THANK YOU

Let's Power a Sustainable
Future Together!

